

Geetha Gurram

Data Engineer

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Professional Summary

Senior Data Engineer with 6.5+ years of experience designing and optimizing scalable data pipelines, ETL/ELT workflows, and cloud-based data platforms, improving data processing efficiency by 35%. Proficient in Python, SQL, Apache Spark, Kafka, and Airflow, with hands-on experience across AWS, Azure, and GCP, reducing data latency by 30%. Experienced in data modeling, data warehousing, and real-time data processing, enabling advanced analytics and machine learning solutions that improved data accessibility by 40%. Skilled at collaborating with cross-functional teams to build reliable, high-performance data architectures, increasing operational efficiency by 25%.

Professional Experience

MetLife, USA

Jan 2025 – Present

Sr. Data Engineer

- Architect and implement scalable data pipelines and ETL/ELT workflows using Python, SQL, Apache Spark, and Apache Airflow, improving data processing efficiency by 45%.
- Design and maintain cloud-native data platforms on AWS (S3, Glue, Lambda, EMR) to support large-scale enterprise analytics and AI workloads.
- Develop real-time streaming pipelines using Apache Kafka and Spark Streaming, reducing data ingestion latency by 35%.
- Implement data quality frameworks, monitoring, and alerting systems, improving pipeline reliability and reducing data failures by 30%.
- Architected optimized data warehouse solutions using Snowflake and Amazon Redshift, slashing data storage costs by 30% through efficient resource allocation using data compression.
- Develop scalable feature engineering pipelines to support machine learning model training and deployment.
- Collaborate with data scientists, analysts, and business stakeholders to deliver scalable data solutions for enterprise analytics and AI-driven applications.
- Implement CI/CD pipelines using Jenkins, Docker, and Kubernetes to automate deployment and improve development productivity by 30%.

Novartis, India

Oct 2022 – Jul 2023

Data Engineer / Generative AI Engineer

- Orchestrated end-to-end data pipelines utilizing Python, Spark, and SQL, elevating data processing speed by 40% and served as the foundation for downstream machine learning models.
- Built LLM-powered document intelligence systems to analyze large volumes of clinical and research data.
- Implemented AI chatbot solutions leveraging LLMs and embeddings, improving internal knowledge access by 35%.
- Designed and implemented Retrieval-Augmented Generation (RAG) architectures using LangChain, OpenAI API, FAISS, and vector databases, improving enterprise knowledge retrieval by 45%.
- Optimized data transformation workflows for structured and unstructured healthcare data sources.
- Championed data quality by designing and deploying 7 automated checks for data integrity, leading to a measurable 30% improvement in data quality, measured across key enterprise datasets.
- Collaborated with data scientists, ML engineers, and research teams to build AI-driven analytics solutions.

Accenture, Hyderabad, India

Oct 2019 – Sep 2022

Big Data Engineer / Data Analyst

- Built and maintained large-scale big data processing pipelines using Apache Spark, Hadoop, and Kafka, processing multi-terabyte datasets.
- Orchestrated a complete overhaul of data warehouse architectures using Snowflake and dimensional data models, accelerating analytics query performance by 40% and enabling real-time reporting.
- Developed scalable data transformation and integration pipelines to ingest data from APIs, relational databases, and enterprise systems.
- Orchestrated the creation of 10+ interactive dashboards leveraging Tableau and Power BI, which catalyzed data-driven insights informing 15-20 key business decisions per quarter across the organization.
- Conducted data analysis and exploratory data analysis (EDA) to identify business trends and performance improvements.
- Pioneered the transition to Apache Airflow for automating data pipeline orchestration, freeing up 10+ engineers daily and increasing the speed of data delivery.
- Worked with cross-functional teams to deliver data-driven insights supporting operations, finance, and customer analytics initiatives.

Virtual Reality K Technologies, Hyderabad, India

Jan 2018 – Sep 2019

Junior Data Engineer / ML Engineer

- Developed and maintained ETL pipelines using Python and SQL, enabling efficient data ingestion and transformation.
- Engineered robust data preprocessing and feature engineering pipelines, increasing model AUC scores by 0.08 and resolving data quality issues that were causing significant model drift.
- Implemented data preprocessing, feature engineering, and model evaluation techniques, improving model performance by 25%.
- Transformed legacy data processing workflows by migrating 7 data pipelines to Apache Spark, accelerating query processing speeds by 45% and boosting insight generation.
- Supported data warehouse development and optimization, improving reporting and analytics capabilities.
- Assisted in building automated reporting systems and dashboards, enabling faster business insights for stakeholders.

Certifications

- Microsoft Certified: Fabric Data Engineer Associate
- Databricks SQL Analytics and BI
- Databricks for generative AI
- Databricks for Data Engineering
- Generative AI Tools
- Getting Started with Autonomous Agents in Copilot Studio

Skills

- **Programming:** Python, SQL, PL/SQL, R, Java, Scala
- **Data Engineering & Big Data::** Apache Spark, PySpark, Kafka, Apache Airflow, ETL/ELT Pipelines, Data Pipelines, Data Modeling, Data Warehousing, Batch & Stream Processing
- **Databases & Data Platforms:** PostgreSQL, MySQL, MongoDB, Oracle, Snowflake, Amazon Redshift, Google BigQuery, Delta Lake
- **Cloud Platforms:** Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)
- **Cloud Data Services:** AWS S3, AWS Glue, AWS Lambda, AWS EMR, Azure Data Factory, Azure Databricks, Azure Synapse Analytics
- **Machine Learning & MLOps:** Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, MLflow, Kubeflow, Feature Engineering, Model Deployment, Model Monitoring
- **Generative AI & LLM Technologies:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases, Embeddings, AI Chatbot Development, LLM Fine-tuning, AI Agents, Document Intelligence Systems
- **Generative AI Tools:** LangChain, LlamaIndex, OpenAI API, Hugging Face, FAISS, Pinecone
- **DevOps & Automation:** Docker, Kubernetes, CI/CD Pipelines, Jenkins, Terraform
- **Data Governance & Security:** Data Privacy, Data Compliance, Data Lineage, Data Catalogs, Access Control, Secure Data Pipelines
- **Data Visualization:** Tableau, Power BI, Matplotlib, Plotly
- **Version Control & Collaboration:** Git, GitHub, GitLab, JIRA, Confluence, Agile (Scrum/Kanban)
- **Professional Skills:** Problem Solving, Analytical Thinking, Cross-Functional Collaboration, Technical Documentation, Stakeholder Communication, Leadership

Education

Master of Science in Information Systems TRINE University, USA	Jan 2024 – May 2025
Bachelor of Technology in Electronics and Communication Engineering G.Pullaiah College of Engineering and Technology, Kurnool, India	May 2013 – Apr 2017